

Engineering Validation & Testing Report: CausalWorks Core Engine

Date: June 4, 2026

Environment: macOS (aarch64-apple-darwin) / Host: lbahl-1

Status: PASSED

1. Executive Summary

This report documents the rigorous local stress-testing and verification of the CausalWorks Replay & Attribution Engine. Moving away from mock environments, the engine's core Rust modules (manifest.rs, ingest.rs, causal.rs) were subjected to a deterministic testing protocol designed to simulate human configuration errors, malformed physical infrastructure data, and edge-case graph topologies.

The software successfully demonstrated fail-closed initialization boundaries, zero-panic zero-allocation parser degradation, and absolute timing precision across millions-fold scale thresholds. The local engine is structurally validated and ready for deterministic digital twin synchronization and offline forensics.

2. Comprehensive Test Ledger & Results

Test 1: The Cyclic Scale Trap (Absurd Unit Validation)

Objective: Verify that the initialization pipeline catches human data-entry mistakes and safely terminates before executing attribution math on invalid data.

Observed Result: INVALID: edge 1->2 has zero or absurd WCET budget 120000000000 ns

Engineering Assessment: PASSED.

Test 2: Destructive Graph Topology Loop (Feedback Attack)

Observed Result: INVALID: dependency cycle detected involving node 1

Engineering Assessment: PASSED.

Test 3: Truncated Payload Resilience (Parser Fuzzing)

Observed Result: frames=5 parsed=4 dropped=1 trips=0 attributed=0 rows=0

Engineering Assessment: PASSED.

Test 4: The Microsecond Horizon (Astronomical Overrun Precision)

Observed Result: overrun=9999500ns (1999900.0%)

Engineering Assessment: PASSED.

Test 5: Clean Non-Timing Fault (The False Catalyst Shield)

Observed Result: UNATTRIBUTED trip — no WCET violation; escalate to fault review.

Engineering Assessment: PASSED.

Test 6: Static Reachability and Orphan Node Lints

Observed Result: Manifest validated with warnings identifying unreachable topology and orphan node configuration.

Engineering Assessment: PASSED.

3. Core Software Architecture Mapping

Raw Stream / PCAP → ingest.rs → manifest.rs → causal.rs → trace.rs

4. Next Milestone Engineering Actions

1. Physical Environment Validation (hwcapture.rs).

2. Digital Twin Synchronization Test using a hardware-in-the-loop or MechLab environment with injected network jitter.